MIL-C-21496D(AS)
5 October 1972

Superseding
MIL-C-21496C(AS)
12 February 1968

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 104 MOD 0

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

This document governs the manufacture, assembly and preparation for delivery of the Cartridge, Impulse, MARK 104 MOD 0 and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-P-223	Powder, Black
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-46610	Primers, Percussion, Styphnate and Chlorate Types, for Small Arms Ammunition

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS AND ASSOCIATED LISTS

Naval Air Systems Command	(Code Ident 30003)
LD 486208	Container, Inner, for Cartridges, Impulse (Formerly Code Ident 10001)
LD 486209	Container, Outer, for Cartridges, Impulse (Formerly Code Ident 10001)
LD 491831	Cartridge, Impulse, MARK 104 MOD 0 (Formerly Code Ident 10001)
1863079	Cartridge, Impulse, MARK 104 MOD 0, Assembly (Formerly Code Ident 10001)
U. S. Army Munitions Command	(Code Ident 19200)
C-7645339	Primer, Percussion (50M-5061W)

PUBLICATIONS

Bureau of Naval Weapons	(Code Ident 10001)
OP 5	Ammunition Ashore (This publication is not furnished to bidders and contractors by the procuring activity. Contractors should contact the Government Inspector for use of the publication.)
WR-43	Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178	Transportation, Explosive and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General.

3.1.1 The MARK 104 MOD 0 impulse cartridge covered by this specification shall be manufactured in accordance with the drawings listed on LD 491831.

3.2 Materials. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referenced on the respective NAVAIR drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy cognizant field activity for cartridges prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 46 MARK 104 MOD 0 impulse cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purposes of this specification, the primer and main propellant charge are defined as primary components. See 4.3.3.

3.4 Primer. The primers shall be furnished from lots manufactured within the preceding 12 months. The primers shall be loaded with Remington Arms Primer Mixture 5061W, manufactured in accordance with Drawing C-7645339; and shall be from lots which were accepted in accordance with the applicable requirements of MIL-P-46610, Class 1. See 4.7.1.

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3.4.1 X-ray. Primers shall be free of the following defects when examined by x-ray in accordance with 4.7.2: missing or abnormally thin charge; charge broken, cracked or porous; cracked or off-center anvil or other misplaced or deformed inert parts; anvil abnormally close to primer cup dome; and all other observable defects in assembly.

3.5 Propellant and ignition charges.

3.5.1 Quantity. All propellant charges and ignition charges shall be as specified on Drawing 1863079. Charges shall be determined by weight rather than by volume measurements.

3.5.2 Black powder. The moisture content of the black powder shall be determined as specified in MIL-P-223, and at the time of loading shall be within the limits specified therein for the applicable class of black powder being used. See 4.7.3.

3.5.3 Smokeless powder, handling and loading. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.5.4 Smokeless powder, condition. The smokeless powder utilized shall exhibit no evidence of decomposition. See 4.7.4.

3.6 Cartridge inspection.

3.6.1 Visual inspection. Cartridges shall be free of the following visible defects: dents, deep scratches, cracks, burrs, split or cracked lip in the crimped area, damaged "O" ring, incomplete or improper sealant application, or any other imperfections which could prevent entry into the firing chamber or adversely affect performance or resistance to moisture. See 4.7.5.

3.6.2 Leakage. Cartridge leakage rate shall not exceed 10^{-5} cc of air or gas per second when tested in accordance with 4.7.6.

3.6.3 X-ray. Cartridges shall be free of imperfections when examined by x-ray in accordance with 4.7.7.

3.7 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.7.8.1, shall meet the requirements listed below when subjected to the test prescribed in 4.7.8.1.

3.7.1 Chamber pressure. The lower (L) limit of maximum chamber pressure shall be 400 pounds per square inch.

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3.7.2 Rate of pressure rise. The upper (U) limit of rate of increase in chamber pressure shall be 10,000 pounds per square inch per second.

3.7.3 Velocity. The upper (U) limit of velocity of the propelled load shall be 31 feet per second.

3.7.4 Misfire. There shall be no misfires. See 4.7.8.2.4.

3.7.5 Hangfire. There shall be no hangfires. See 4.7.8.2.5.

3.8 Change in propellant lot. Whenever a change occurs in the propellant lot for the main charge, test firings shall be conducted in accordance with 4.10.

3.9 Packaging, packing and marking. The packaging and packing of cartridges and the container markings shall be in accordance with Section 5. See 4.6.1.

3.10 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be directed to dimensions, finishes and coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105 or MIL-STD-414, whichever applies.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation and size shall be in accordance with MIL-STD-105 or MIL-STD-414, whichever applies.

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4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 46 MARK 104 MOD 0 impulse cartridges shall be prepared. Of these cartridges, 36 shall be expended in the tests listed below and ten shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.3 through 3.6.3 of this specification. Preproduction sample acceptance shall consist of the following tests as prescribed in MIL-D-21625: 15 "g" shock (6); vibration (6); temperature and humidity cycling (12). Subsequent to environmental treatment, these cartridges shall be conditioned for a minimum of six hours at +70° ±5°F and fired in the test set of 4.7.8.1. The numbers in parentheses denote the allocation of cartridges for the tests. In addition, 12 cartridges, not subjected to environmental treatment, shall be conditioned for a minimum of six hours at +70° ± 5°F and fired in the test set of 4.7.8.1. All cartridges shall fire and produce results within the limits set forth in 3.7.1, 3.7.2, 3.7.3 and 3.7.5. The failure of any cartridge to meet the acceptance requirements as stated above shall be cause for rejection of the preproduction sample.

4.3.3 Production lot. Unless otherwise specified in the contract, the minimum and maximum production lot size shall be 801 and 1300 cartridges, respectively. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.4 Gages. The contractor shall provide gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C), and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Quality conformance inspection. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of LD 491831, and verification of the following requirements and tests:

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Item	Nature of Requirement Verification	Applicable Paragraphs	Classification (WR-43)
Primary component	Single lot	3.4, 4.3.3	(M101)
Primer	Age, loading and acceptance	3.4, 4.7.1	(M102)
Primer	X-ray	3.4.1, 4.7.2	(M103)
Propellant and ignition charges	Type and weight	3.5.1	(M104)
Black powder	Moisture content	3.5.2, 4.7.3	(M105)
Smokeless powder	Condition	3.5.4, 4.7.4	(M106)
Cartridge	Visual inspection	3.6.1, 4.7.5	(M107)
Cartridge	Leakage	3.6.2, 4.7.6	(M108)
Cartridge	X-ray	3.6.3, 4.7.7	(M109)
Cartridge	Ballistic test	3.7, 4.7.8	(M110)
Cartridge	Ballistics (change in propellant lot)	3.8, 4.10	(M111)
Cartridge	Packaging, packing and container markings	3.9, 4.6.1	(M112)

4.6.1 Packaging, packing and marking. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in Section 5.

4.7 Tests.

4.7.1 Primer verification. The contractor shall provide verification that the primers to be used were manufactured, loaded and accepted in accordance with the requirements of 3.4 and were furnished from lots manufactured within the preceding 12 months.

4.7.2 Primer x-ray. Examine all primers by x-ray and reject those having any of the defects of 3.4.1. For the most revealing exposure, the primers shall be laid on their sides with the anvil cutouts vertical. Pressure sensitive tape may be used as an aid in positioning the primers.

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4.7.3 Moisture content of black powder. Prior to the loading of each production lot and daily prior to loading any cartridges, whether or not for a new production lot, the black powder to be used shall be tested for moisture content in accordance with 4.7.3.1. Additional tests shall be conducted when such tests are considered warranted by variations in conditions or process which could increase the moisture content of the powder. The presence of moisture in excess of the limits specified in MIL-P-223 shall cause suspension of loading activities until corrective measures have been taken.

4.7.3.1 Determining moisture content. At the site of loading, take from the powder container a sample of approximately two grams of the black powder to be used, place in a tared container and accurately determine the net weight of the sample to the nearest 0.1 milligram. Dry the sample for a minimum of 90 minutes at $90^{\circ} \pm 5^{\circ}C$. Cover the container, cool and accurately determine the net weight of the sample to the nearest 0.1 milligram. The weight loss is calculated as the percentage of moisture in the sample prior to drying.

4.7.4 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

4.7.5 Visual inspection. Examine all completely assembled cartridges and reject those having any of the defects noted in 3.6.1.

4.7.6 Leakage. Each cartridge in the preproduction sample or production lot ballistic sample shall be leak tested in a dry gas or dry air leak tester. Cartridges which exhibit a leak rate in excess of 10^{-5} cc of air or gas per second at a pressure differential of 1.0 ± 0.1 atmosphere shall be considered defective. Cartridges of the preproduction sample which exhibit a leakage defect shall be rejected. If one or more cartridges of the production lot sample exhibit a leakage defect, then a 100 percent leak test of the lot being tested shall be required at no additional expense to the Government. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected.

4.7.7 X-ray examination. Examine all cartridges by x-ray and reject those having any observable imperfections in assembly. Particular attention shall be given to the presence of the propellant charge.

4.7.8 Ballistic tests.

4.7.8.1 The number of cartridges selected from each production lot for the ballistic tests shall be in accordance with the requirements of Table A-2 of MIL-STD-414, Inspection Level IV and an AQL of 0.04. The cartridges shall be conditioned for a minimum of six hours at $70^{\circ} \pm 5^{\circ}F$ and fired in a 6EQ2 type of training tower catapult. The catapult

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firing chamber shall be thoroughly cleaned before each firing. Sufficient weight shall be added to the propelled catapult seat to provide a total propelled load of 300 ± 3 pounds. Suitable instrumentation shall be used for each firing to record (1) the elapsed time between actuation of the firing mechanism and the start of pressure rise in the firing chamber, (2) firing chamber pressure versus time, and (3) velocity of the propelled load. In addition to the sample of cartridges for the above test, ten additional cartridges shall be furnished from each production lot for investigative purposes.

4.8.3.2 Criteria of acceptability.

4.7.8.2.1 Chamber pressure. The lower (L) limit of maximum chamber pressure shall be as specified in 3.7.1. Lot acceptability shall be in accordance with the applicable provisions of MIL-STD-414 at an AQL of 0.04.

4.7.8.2.2 Rate of pressure rise. The rate of pressure rise shall be determined from the pressure versus time record by measuring the increase in pressure over a period of 30 milliseconds. The measurement shall be made where the rate of pressure rise is maximum and essentially linear, neglecting small irregularities, over the period of measurement. The upper (U) limit of rate of increase in chamber pressure shall be as specified in 3.7.2. Lot acceptability shall be in accordance with the applicable provisions of MIL-STD-414 at an AQL of 0.04.

4.7.8.2.3 Velocity. The velocity of the propelled load shall be measured over an interval of 6.00 ± 0.06 inches commencing 21.00 ± 0.12 inches from the initial position of the load. The upper (U) limit of velocity of the propelled load shall be as specified in 3.7.3. Lot acceptability shall be in accordance with the applicable provisions of MIL-STD-414 at an AQL of 0.04.

4.7.8.2.4 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.7.8.2.5 Hangfire. One or more hangfires shall result in rejection of the lot represented. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and the start of pressure rise exceeds 0.050 second.

4.8 Retest. There shall be no retests.

4.9 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and that part of the test reconducted.

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4.10 Change in propellant lot. Whenever a change in propellant lot occurs for the main propellant charge, validity of the charge determination shall be verified by firing 12 cartridges which have been temperature conditioned for a minimum of six hours at +70° ± 5°F. Results of these firings shall fall within the limits prescribed in 3.7.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. One cartridge shall be packaged in an inner container in accordance with the drawings listed on LD 486208.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be packaged in accordance with Method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 171-178.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Twelve inner containers as described in 5.1.1 shall be packed in accordance with the drawings listed on LD 486209.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring activity, cartridges intended for expenditure in preproduction sample testing and for production lot acceptance tests and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3.2 Normal markings. Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers

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shall be in accordance with the drawings listed on LD 486208 and LD 486209, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 Intended use. The MARK 104 MOD 0 impulse cartridge was originally designed only for use in the 6EQ2 training tower catapult.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required and, if so, specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order.

6.3.1 Test report. The contract or order should specify that results of ballistic tests are to be forwarded to the Naval Air Systems Command (AIR-53222A) and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. For the purposes of this specification, the following definitions apply.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

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6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 Explosives safety precautions. Safety precautions for explosive-loaded items are contained in DOD 4145.26M, "DOD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material".

Project No. 1377-N511

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-21496D(AS) CARTRIDGE, IMPULSE MARK 104 MOD 0

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE NAVY
Naval Air Systems Command
Washington, D.C. 20360

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